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The Editor
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Dear Editor,

I am submitting the manuscript entitled “Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Decomposition of Size-and-Surface Factor Sensitivity Under a Common ISO Stress Baseline” for consideration for publication in the *International Journal of Fatigue*.

This paper addresses a practical problem in gear design: when three qualified engineers compute the bending fatigue life of the same spur gear—identical geometry, material, and load—their predictions can span five orders of magnitude. The divergence originates not from material variability but from the methodological choices each engineer legitimately makes under the letter of published standards.

The manuscript makes three contributions:

1. **First full-factorial variance decomposition across five simultaneous engineering choices in gear fatigue.** No prior study has ranked the contributions of standard selection, mean-stress correction method, surface roughness grade, S–N curve source, and cumulative damage rule within a single controlled experimental design. The analysis employs a censored-likelihood AFT framework that preserves information from the 28 run-out observations (19% of the 144-combination dataset)—unlike conventional ANOVA or Sobol’ sensitivity analysis, which must discard them.
2. **Demonstration that method-induced uncertainty dominates the prediction budget.** A quantitative benchmark against published 42CrMo4 gear fatigue test data (Wang et al., 2023) shows that the method-induced $\log_{10}(N_f)$ spread (2.1–3.3 dex) is 2–3 times larger than the intrinsic material scatter (~ 1 dex) at a fixed stress level. The three dominant factors (mean-stress correction, size-and-surface factor formulation, surface roughness) together account for 88.6% of total log-variance.
3. **Translation of numerical results into actionable engineering guidance.** The paper provides cross-standard empirical correction formulas, a design risk map ranking methodological choices by conservatism direction, consequence-classified uncertainty thresholds, and a step-by-step design protocol for controlling prediction scatter.

A complete reproducibility package is submitted as Supplementary Material, including all Python source code, the 144-combination sweep data, and a one-click reproduction script (`bash reproduce.sh`).

This manuscript has not been published previously and is not under consideration elsewhere. All authors have approved the submission. The author declares no competing interests. Generative AI tools (Claude/Anthropic) were used for language editing and code development; all scientific content, analysis decisions, and numerical results are the author's own.

I believe this work fits the scope of IJF as a computational uncertainty quantification study that addresses a recognized practical problem in gear fatigue assessment. I look forward to the reviewers' evaluation.

Sincerely,

Zhilei Chen